EE 445/645: Physical Models in Remote Sensing (Spring 2026)

Chapter 03-Part 01:Radiative Quantities of Interest– Notes (See C3-P1 PPTs)

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Slide 1: Title – Physical Models in Remote Sensing (Chapter 03, Part 01)

Overview

This lecture series, Chapter 03 – Part 01, introduces the radiative transfer (RT) framework used to model how electromagnetic radiation interacts with vegetation canopies in the context of optical remote sensing. The overarching goal is to derive physically meaningful quantities that can be measured by satellite or airborne sensors and used to infer vegetation properties.

Topics Covered in This Chapter

The chapter covers the following interconnected themes:

- Turbid Medium Approximation – the conceptual idealization of a vegetation canopy
- Simulated Temperate Forest – a worked example with real structural parameters
- Assumptions – the simplifying conditions that make the RT problem tractable
- The Horizontally Homogeneous Medium – the 1D canopy geometry
- Solar and Sky Illumination – the two components of the incoming radiation field
- The Incident Radiation Field – mathematical description of illumination
- 1D RT Problem – formulation of the radiative transfer equation
- Solution – interpretation of what a solved RT problem provides
- Radiative Quantities of Interest – the suite of quantities retrievable from the solution
- G-Function in Coniferous Canopies – a structural quantity governing radiation interception

Why It Matters

Physical models in remote sensing translate the abstract physics of light–canopy interaction into measurable signals. Understanding these models is essential for interpreting satellite products (e.g., MODIS surface reflectance, albedo, FPAR) and for developing retrieval algorithms that estimate biophysical variables such as leaf area index (LAI) or canopy water content.

Slide 2: Turbid Medium Approximation

The Core Challenge

Real vegetation canopies — forests, crops, shrublands — are extraordinarily complex: leaves vary in size, shape, angle, and spatial arrangement; branches, stems, and woody elements add structural clutter; and the canopy changes continuously through growth and senescence. Fully representing this complexity in a radiation model is computationally prohibitive and often unnecessary for the purposes of large-scale remote sensing.

The Turbid Medium Concept

The turbid medium (or 'green gas') approximation treats the canopy as a continuous, homogeneous medium densely filled with infinitely small, planar leaf elements of negligible thickness and area. This is analogous to how atmospheric scattering is treated in radiative transfer theory — the discrete individual molecules are replaced by a continuous scattering medium described by bulk optical properties.

- The medium is characterized by two key variables: (1) Leaf Area Density (LAD) — the one-sided leaf area per unit canopy volume [m^2 m^{-3}]; (2) Leaf Normal Orientation Distribution — the statistical distribution of leaf orientations (e.g., spherical, planophile, erectophile distributions)
- All non-leaf elements (branches, trunks, stems) are ignored. This is justified because leaves dominate the radiative exchange and photosynthesis within the canopy.
- The abstraction captures the most radiatively significant aspect of the canopy: the interception and scattering of solar radiation by green foliage.

Ecological Significance

Despite its simplicity, the turbid medium approximation reproduces many observed canopy reflectance and transmittance characteristics with good accuracy, especially for broadleaf and mixed canopies with moderate-to-high LAI. It is the foundation of virtually all 1D canopy reflectance models, including SAIL, PROSAIL, and others widely used in remote sensing.

Slide 3: Simulated Temperate Forest – Lidar Cloud of Foliage

Purpose of This Slide

This slide bridges the gap between the abstract turbid medium concept and real-world canopy structure by showing a simulated Lidar point cloud of a temperate broadleaf deciduous forest. It illustrates the kind of structural complexity that the turbid medium approximation seeks to capture.

Simulation Parameters

- Plot dimensions: 120 × 60 m
- Tree density: 500 individual trees (~700 trees/ha), randomly located
- Species composition: 10 broadleaf deciduous species
- Leaf Area Index (LAI): 3.84 $m^2\ m^{-2}$ (total one-sided leaf area per unit ground area)
- Individual leaf area: ~0.041 m^2 per leaf
- Point cloud size: 750,000–850,000 points total; ~700,000 are leaf points; the remainder represent branches and other woody foliage elements

Key Concepts

Leaf Area Index (LAI) is one of the most important biophysical variables in remote sensing and ecology. It directly controls canopy light interception, transpiration, and carbon assimilation. An LAI of ~3.84 is typical of temperate deciduous forests in summer. The simulation demonstrates how the turbid medium's 'leaf area density' parameter translates into a realistic spatial distribution of foliage. Lidar (Light Detection and Ranging) is an active remote sensing technique that measures distances by timing laser pulses — the resulting point clouds can be used to directly measure canopy structure and validate turbid medium models.

Slide 4: Assumptions of the Standard RT Problem

Why Assumptions Matter

Every physical model rests on assumptions that define the scope of its applicability. The standard 1D radiative transfer problem for vegetation canopies is built on a well-defined set of assumptions that simplify the mathematics while preserving the physics relevant to optical remote sensing.

List of Assumptions

- Horizontally homogeneous canopy: Canopy properties (LAD, leaf orientation) vary only with depth (z), not horizontally. This reduces the problem from 3D to 1D.
- Planar leaves only: The medium consists exclusively of flat leaf elements. Non-photosynthetic elements (branches, stems) are excluded.
- Illumination from above only: The canopy is illuminated by (a) direct solar radiation and (b) isotropic diffuse sky radiation — both entering from the top boundary.
- Stationary (time-independent) problem: No temporal variation is modeled. The radiation field is in steady state.
- No frequency shifting: There is no fluorescence or Raman scattering — each photon maintains its wavelength throughout interactions.
- Wavelength-dependent scattering: Leaf reflectance and transmittance vary with wavelength (λ), so the problem is solved independently at each wavelength.
- No emission within the medium: At solar wavelengths, thermal emission by leaves is negligible and is excluded.
- Non-reflecting top and bottom boundaries: Incoming radiation is not pre-reflected at the canopy top; the soil boundary at the bottom is treated as non-reflecting in the standard formulation (can be relaxed).
- Non-re-entrant medium: Radiation exiting through the bottom boundary does not re-enter the canopy from below.

Implications

These assumptions collectively define what is called the 'standard problem' in optical remote sensing of vegetation. Each assumption can, in principle, be relaxed at the cost of greater mathematical and computational complexity. For example, including soil reflectance at the lower boundary (as in SAIL) significantly improves model realism for sparse canopies.

Slide 5: Schematic – Horizontally Homogeneous Medium

The 1D Canopy Geometry

This slide introduces the coordinate system and geometric structure of the horizontally homogeneous (1D) canopy model. The canopy is represented as a plane-parallel slab bounded by two horizontal planes.

Geometry Description

- The z-axis is oriented vertically downward (or from top to bottom of the canopy, depending on convention).
- $z = 0$ marks the top boundary — the interface between the atmosphere and the canopy top.
- $z = z_H$ marks the bottom boundary — the soil surface or the base of the canopy.
- Between these two boundaries lies the leaf canopy, a medium with spatially varying (but horizontally uniform) leaf area density and leaf orientation distribution.
- The horizontal extent is effectively infinite — there are no edge effects.

Mathematical Significance

The horizontal homogeneity assumption means that all radiative quantities (intensity, flux) depend only on depth z and direction Ω, not on horizontal position (x, y). This reduces the three-dimensional radiative transfer equation to a one-dimensional integro-differential equation, which is far more tractable analytically and numerically. This geometry is identical to that used in atmospheric radiative transfer (e.g., for cloud or ozone layer calculations), allowing the same mathematical tools to be applied.

Slide 6: Schematic – Solar and Sky Illumination

Two Components of the Incident Radiation Field

The top boundary of the canopy is illuminated by two distinct components of incoming solar radiation. Their different angular distributions are the key distinction between them, and both must be accounted for in realistic RT models.

Direct Solar Radiation

- Symbol: $I_0\lambda \cdot \delta(\Omega - \Omega_0)$
- Characterized by a single direction Ω_0 — the direction from the sun to the canopy.
- The Dirac delta function $\delta(\Omega - \Omega_0)$ expresses that all energy arrives from exactly one direction.
- This component casts geometric shadows within the canopy, producing the 'hot spot' effect (enhanced reflectance in the backscatter direction).
- Its intensity $I_0\lambda$ varies with wavelength and solar position.

Diffuse Sky Radiation

- Symbol: $Id\lambda$ (integrated over the upper hemisphere Ωd)
- Originates from scattering of sunlight by the atmosphere — clouds, aerosols, and molecular (Rayleigh) scattering all contribute.
- Treated as isotropic (uniform in all downward directions) in the standard problem, though in reality the sky radiance distribution is not perfectly isotropic.
- On overcast days, diffuse sky radiation dominates and the canopy illumination is essentially Lambertian.

Practical Importance

The ratio of diffuse to total irradiance (the diffuse fraction) varies dramatically with atmospheric conditions (clear sky: ~10–20%; overcast: ~100%). Because direct and diffuse radiation interact differently with the canopy, satellite-derived surface reflectance products must account for both components. MODIS albedo products, for instance, are reported separately for purely direct (black-sky albedo / DHR) and purely diffuse (white-sky albedo / BHR) illumination.

Slide 7: The Incident Radiant Field – Mathematical Description

Formalizing the Illumination

This slide provides the mathematical expressions for the two illumination components — direct solar and diffuse sky — that enter at the top of the canopy. These expressions are the boundary conditions for the RT equation.

Direct Solar Radiation

- Intensity: $I_{o\lambda} \cdot \delta(\Omega - \Omega_o)$ — a spectrally dependent intensity concentrated in direction Ω_o.
- Flux Density (irradiance): $F_{o\lambda} = |\mu_o| \cdot I_{o\lambda}$, where $\mu_o = \cos(\theta_o)$ is the cosine of the solar zenith angle θ_o.
- The factor $|\mu_o|$ accounts for the geometric projection of the solar beam onto the horizontal surface — at high solar zenith angles (low sun), the same beam energy is spread over a larger horizontal area, reducing the flux density. This is the familiar 'cos θ' factor in solar energy calculations.

Diffuse Sky Radiation

- Intensity: $I_{d\lambda}$ — assumed constant (isotropic) across all downward directions.
- Flux Density: $F_{d\lambda} = \pi \cdot I_{d\lambda}$ — obtained by integrating the isotropic radiance over the downward hemisphere.
- The factor π emerges from the solid-angle integral $\int_{2\pi^-} |\mu_d|\, d\Omega_d = \pi$ for isotropic radiance. This is a standard result in radiometry.

Units and Radiometric Conventions

Intensity (radiance) has units of $W\ m^{-2}\ sr^{-1}\ nm^{-1}$. Flux density (irradiance) has units of $W\ m^{-2}\ nm^{-1}$. The subscript λ denotes per-unit-wavelength quantities (spectral). These distinctions are critical for correctly interpreting and comparing remote sensing measurements.

Slide 8: The 1D Monochromatic Radiative Transfer (RT) Problem

Problem Statement

This slide presents the full formulation of the 1D monochromatic radiative transfer problem for a vegetation canopy. 'Monochromatic' means the problem is solved at a single wavelength λ; in practice, it must be solved at every wavelength of interest. The schematic integrates all elements introduced so far: geometry, illumination (direct + diffuse), and the medium.

Inputs to the RT Problem

- Canopy structure: Leaf area density $u(z)$ and leaf normal orientation distribution $g(\Omega_L)$ as functions of depth z.
- Leaf optical properties: Leaf reflectance $\rho_l\lambda$ and transmittance $\tau_l\lambda$ at wavelength λ.
- Boundary conditions at $z = 0$: Direct solar intensity $I_0\lambda\ \delta(\Omega - \Omega_0)$ and diffuse sky intensity $Id\lambda$.
- Boundary conditions at $z = z_H$: Non-reflecting lower boundary (or soil reflectance if specified).

The RT Equation

The 1D RT equation expresses how the intensity field $I(z, \Omega)$ changes with depth z due to: (1) Extinction — attenuation of radiation by leaf interception; (2) Scattering — redirection of intercepted radiation into other directions by leaf reflection and transmission. The equation is an integro-differential equation because the source term (scattered radiation) involves an integral over all directions. The key structural parameter controlling extinction is the G-function $G(z, \Omega)$, which describes the mean projection of leaf area onto a plane perpendicular to direction Ω.

Solution Strategy

Because of the Dirac delta illumination term, it is standard to split the intensity into uncollided (direct solar, uncollided diffuse sky) and collided (scattered) components and solve for each separately. The uncollided components have analytical solutions (Beer–Lambert law); the collided component requires numerical methods or further approximations.

Slide 9: Solution of the RT Problem

What 'Solved' Means

This slide clarifies what it means to have a 'solution' to the RT problem. Solving the RT equation yields the complete radiation field throughout the canopy — both at every depth z and in every direction Ω. This is the fundamental output from which all practical radiometric quantities are derived.

Components of the Solution

- Uncollided direct solar intensity: $I_{0\lambda}(z, \Omega_0)$ — the unscattered solar beam at depth z, propagating in the solar direction Ω_0. This decays with depth following Beer–Lambert attenuation: $I_{0\lambda}(z) = I_{0\lambda} \cdot \exp(-G(\Omega_0) \cdot u(z) / |\mu_0|)$, where u(z) is the cumulative leaf area index from the top.
- Uncollided diffuse sky intensity: $I_{d\lambda}(z, \Omega_d)$ — the unscattered diffuse sky radiation at depth z in each downward direction Ω_d. Also attenuated by Beer–Lambert law.
- Collided (scattered) intensity: $I_{s\lambda}(z, \Omega)$ for all directions Ω (upward and downward) — radiation that has been scattered one or more times by leaves within the canopy. This is the most complex component and the one that gives rise to the canopy reflectance signal observed by sensors.

Key Insight

The separation into uncollided and collided components is physically meaningful and computationally efficient. The collided component $I_{s\lambda}(z, \Omega)$ is what carries information about canopy structure and leaf optical properties in the upward direction toward the sensor. It is the quantity from which all directional reflectance products (HDRF, BRF, BRDF) are derived.

Slide 10: Radiative Quantities of Interest – Overview

Why These Quantities?

Once the RT problem is solved, we can compute a range of radiometric quantities that are directly measurable by satellite sensors or relevant to ecological and energy-balance applications. This slide organizes these quantities into three categories based on their angular character.

Category A: Basic Radiative Fluxes

- Reflectivity (FR): Fraction of incident flux reflected upward by the canopy.
- Transmissivity (FT): Fraction of incident flux transmitted downward through the canopy.
- Absorptivity (FA): Fraction of incident flux absorbed within the canopy.
- These three must sum to unity (energy conservation): FR + FT + FA = 1.

Category B: Flux-Based Remote Sensing Products

- Bi-Hemispherical Reflectance (BHR) — also called white-sky albedo
- Directional-Hemispherical Reflectance (DHR) — also called black-sky albedo
- Albedo — spectrally integrated BHR
- FPAR — Fraction of Absorbed Photosynthetically Active Radiation (400–700 nm)

Category C: Directional Remote Sensing Products

- Hemispherical-Directional Reflectance Factor (HDRF)
- Bi-Directional Reflectance Factor (BRF)
- Bi-Directional Reflectance Distribution Function (BRDF)

Practical Context

Categories B and C correspond directly to standard products delivered by operational satellite missions such as MODIS, Sentinel-2, and Landsat. Understanding their precise definitions is essential for correctly interpreting satellite data and for inter-comparing products from different sensors.

Slide 11: Category A – Reflectivity, Transmissivity, and Absorptivity

Definitions

The three bulk radiative flux ratios describe how the total incident energy is partitioned by the canopy system.

Formulas

- Reflectivity $FR(\lambda)$ = Reflected flux / Total incident flux = $[\int_{2\pi^+} |\mu|\, I_{s\lambda}(z=0, \Omega)\, d\Omega] / (F_{0\lambda} + F_{d\lambda})$
- Transmissivity $FT(\lambda)$ = Transmitted flux / Total incident flux = [Flux exiting at $z=z_H$] / $(F_{0\lambda} + F_{d\lambda})$
- Absorptivity $FA(\lambda)$ = Absorbed flux / Total incident flux = $[\int_0^{z_H}$ Absorbed flux at z dz$]$ / $(F_{0\lambda} + F_{d\lambda})$
- Energy conservation: FR + FT + FA = 1 (this provides a useful check on the accuracy of a numerical solution)

Important Notes

- All three quantities are dimensionless (ratios).
- All implicitly depend on solar direction Ω_0 because the uncollided components of the transmitted and absorbed flux depend on the angle of the solar beam.
- They also depend on the ratio of direct to diffuse illumination ($F_{0\lambda}$ vs. $F_{d\lambda}$).
- These are integrated (hemispherical) quantities — they contain no information about the direction from which an observer views the canopy.

Slide 12: Canopy Reflectivity

Definition

Canopy reflectivity $FR\lambda$ at the top boundary ($z = 0$) is defined as the ratio of the upward-scattered flux to the total incident (downward) flux.

Formula

$FR\lambda(z=0) = [\int_{2\pi^+} |\mu|\ Is\lambda(z=0, \Omega)\ d\Omega] / (F_0\lambda + Fd\lambda)$

- Numerator: Hemispherical upward flux at z=0, obtained by integrating the collided intensity $Is\lambda$ weighted by the cosine $|\mu|$ over the upward hemisphere ($2\pi^+$).
- Denominator: Total downward incident flux = direct solar flux $F_0\lambda$ plus diffuse sky flux $Fd\lambda$.
- The integration $\int_{2\pi^+} |\mu|\ Is\lambda\ d\Omega$ is a standard flux (irradiance) calculation: radiance multiplied by the Lambert cosine factor and integrated over solid angle.

Physical Interpretation

Reflectivity is the 'bulk mirror efficiency' of the canopy — what fraction of all incoming energy is returned upward. It depends on leaf optical properties (ρ_l, τ_l), LAI, leaf orientation, and illumination geometry. For a dense canopy with high LAI, reflectivity is primarily determined by leaf reflectance in the visible and by multiple scattering in the near-infrared.

Slide 13: Canopy Transmissivity

Definition

Canopy transmissivity $FT\lambda$ is the ratio of total downward flux exiting at the canopy bottom ($z = z_H$) to the total incident flux at the top.

Three Components of Transmitted Flux

- Component A — Uncollided direct solar radiation: $A = |\mu_0| \cdot I_0\lambda(z=z_H)$. This is the unscattered solar beam that has penetrated through all the gaps in the canopy without any leaf interaction. It follows Beer–Lambert attenuation.
- Component B — Uncollided diffuse sky radiation: $B = \int_{2\pi^-} |\mu_d|\ I_d\lambda(z=z_H, \Omega_d)\ d\Omega_d$. The fraction of diffuse sky radiation that has passed through the canopy without scattering.
- Component C — Collided (scattered) radiation: $C = \int_{2\pi^-} |\mu|\ I_s\lambda(z=z_H, \Omega)\ d\Omega$. Radiation that has been scattered one or more times by leaves and is exiting downward at the bottom.

Key Points

- The distinction between uncollided and collided transmitted radiation is important: uncollided transmission determines the sunfleck pattern on the forest floor, which is ecologically significant for understory photosynthesis.
- Gap fraction — the probability that a ray in direction Ω passes through the canopy without interception — is directly related to Component A of the transmissivity for the solar direction.
- Energy conservation check: $FR\lambda + FT\lambda + FA\lambda = 1$.

Slide 14: Canopy Absorptivity

Definition

Canopy absorptivity $F_{A\lambda}$ is the ratio of the total radiation absorbed throughout the canopy volume to the total incident flux. It requires integrating absorbed flux contributions over all depths z.

Three Absorption Components

- $F_{0A\lambda}(z)$ — absorbed uncollided direct solar radiation at depth z: $F_{0A\lambda}(z) = |\mu_0| \cdot I_{0\lambda}(z) \cdot \sigma_{a\lambda}(z, \Omega_0)$, where $\sigma_{a\lambda}$ is the absorption cross-section per unit volume (related to leaf absorption coefficient and LAD).
- $F_{dA\lambda}(z)$ — absorbed uncollided diffuse sky radiation at depth z: $F_{dA\lambda}(z) = \int_{2\pi^-} |\mu_d|\, I_{d\lambda}(z, \Omega_d)\, \sigma_{a\lambda}(z, \Omega_d)\, d\Omega_d$. Integration over all downward directions of sky radiation.
- $F_{sA\lambda}(z)$ — absorbed collided (scattered) radiation at depth z: $F_{sA\lambda}(z) = \int_{4\pi} |\mu|\, I_{s\lambda}(z, \Omega)\, \sigma_{a\lambda}(z, \Omega)\, d\Omega$. The integration is over the full sphere (4π) because scattered radiation travels in all directions.

Ecological Significance

Absorptivity is closely related to FPAR (see Slide 19) when restricted to the photosynthetically active radiation (PAR) spectral band. Absorbed radiation is the energy currency of photosynthesis and surface energy balance. The (1 − Albedo) fraction of total absorbed solar energy drives sensible and latent heat fluxes, making this quantity central to climate modeling.

Slide 15: Category B – Radiative Fluxes of Interest in Remote Sensing

Overview

Category B quantities are spectrally and/or hemispherically integrated reflectance metrics that are directly retrieved from satellite sensors and delivered as operational products. They characterize the canopy's overall 'brightness' under various illumination conditions.

The Four Category B Quantities

- Bi-Hemispherical Reflectance (BHR): Reflectivity under both direct solar and diffuse sky illumination. Also called white-sky albedo.
- Directional-Hemispherical Reflectance (DHR): Reflectivity under solar illumination only (no diffuse sky). Also called black-sky albedo.
- Albedo: BHR integrated over the full solar spectrum (broadband).
- FPAR: Fraction of incident PAR (400–700 nm) that is absorbed by the canopy.

MODIS Products

All four quantities listed here are produced routinely by the MODIS instrument aboard NASA's Terra and Aqua satellites. MODIS delivers global maps of BHR, DHR, Albedo, and FPAR at 500 m and 1 km resolution, updated every 8 or 16 days. These products are widely used in climate models, land surface schemes, and carbon cycle studies.

Slide 16: Bi-Hemispherical Reflectance (BHR) – White-Sky Albedo

Definition

BHR is the canopy reflectivity under combined solar and sky illumination. It is numerically identical to the canopy reflectivity $FR\lambda$ defined in Slide 12.

Formula

$BHR\lambda = FR\lambda = [\int_{2\pi^+} |\mu|\ Is\lambda(z=0, \Omega)\ d\Omega] / (F_0\lambda + Fd\lambda)$

- Numerator: Upward hemispherical flux of scattered radiation at the canopy top.
- Denominator: Total incident flux (direct + diffuse).
- The 'bi-hemispherical' label refers to the fact that illumination comes from the full downward hemisphere (both direct and diffuse) and the reflected flux is integrated over the full upward hemisphere.

White-Sky Albedo

The term 'white-sky albedo' is used because BHR corresponds to the reflectance under diffuse (isotropic) illumination

MODIS Product

BHR is a standard MODIS product (Collection 6/6.1, product MCD43). It is derived through BRDF model inversion from multi-angle reflectance observations acquired over 16-day compositing periods.

Slide 17: Directional-Hemispherical Reflectance (DHR) – Black-Sky Albedo

Definition

DHR is the canopy reflectivity under direct solar illumination only — the diffuse sky term $F_{d\lambda}$ is excluded from both the incident flux and the scattering source terms.

Formula

$DHR_\lambda = [\int_{2\pi^+} |\mu|\ I_{s\lambda}(z=0, \Omega)\ d\Omega] / F_{0\lambda}$

- Numerator: Upward hemispherical flux at z=0, now driven only by direct solar illumination.
- Denominator: Direct solar flux $F_{0\lambda}$ only (no diffuse term).
- DHR depends on solar zenith angle θ_0 (i.e., on Ω_0), unlike BHR which averages over all illumination directions.

Black-Sky Albedo

The term 'black-sky albedo' is used because DHR corresponds to the reflectance under a completely black sky — only the solar disk illuminates the surface, and there is no diffuse sky radiation at all. This condition is approximated under very clear, dry-atmosphere conditions. It represents the other end-member of realistic albedo.

Operational Use

Under actual atmospheric conditions, the true surface albedo lies between BHR (white-sky) and DHR (black-sky), weighted by the diffuse fraction of irradiance: Albedo_actual ≈ (1−f_d) · DHR + f_d · BHR, where f_d is the diffuse fraction. This formula is used in land surface models to compute actual surface albedo from MODIS products.

Slide 18: Canopy Albedo

Definition

Canopy Albedo is the spectrally broadband version of reflectivity. It is computed by integrating $FR\lambda$ (the spectral reflectivity) over the entire solar spectrum.

Formula

Albedo = $[\int_{\lambda_1}^{\lambda_2} FR\lambda \, d\lambda] / [\int_{\lambda_1}^{\lambda_2} (F_0\lambda + Fd\lambda) \, d\lambda]$

- λ_1 and λ_2 are the wavelength limits of solar radiation (approximately 300–4000 nm for shortwave).
- The numerator integrates reflected spectral flux over the solar spectrum.
- The denominator integrates total incident spectral flux over the same range.
- Result is dimensionless.

Energy Balance Importance

Since (1 − Albedo) gives the fraction of total solar radiation absorbed by the surface, albedo is a critical parameter in surface energy balance: Net Solar Radiation = (1 − Albedo) × Incoming Solar Radiation. Variations in albedo drive significant changes in surface temperature and regional climate. For example, deforestation typically increases albedo (bright bare soil replacing dark forest), which has a cooling effect that partially offsets the warming from released carbon.

MODIS Product

MODIS provides broadband shortwave albedo as part of the MCD43 product suite, at 500 m spatial resolution and 8-day temporal resolution.

Slide 19: Fraction of Absorbed PAR (FPAR)

Definition

FPAR is the fraction of incident Photosynthetically Active Radiation (PAR) that is absorbed by the vegetation canopy. PAR is defined as the 400–700 nm spectral range — the wavelengths of light that can drive photosynthesis.

Formula

$FPAR = [\int_{400}^{700} FA\lambda \, d\lambda] / [\int_{400}^{700} (F_0\lambda + Fd\lambda) \, d\lambda]$

- Numerator: Total absorbed PAR flux, obtained by spectrally integrating the absorbed spectral flux $FA\lambda$ over 400–700 nm.
- Denominator: Total incident PAR flux (direct + diffuse) over 400–700 nm.
- FPAR is dimensionless and ranges from 0 (no absorption) to 1 (complete absorption).

Carbon Cycle Significance

FPAR is a key driver of the Monteith model of Gross Primary Production (GPP): $GPP = FPAR \times PAR \times \varepsilon^*$, where ε^* is the light-use efficiency (mol CO_2 per mol absorbed PAR photons). This simple relationship connects satellite-observable FPAR directly to ecosystem carbon uptake, making it one of the most ecologically important variables in remote sensing. It is also a direct input to most land surface models and dynamic global vegetation models (DGVMs).

MODIS Product

FPAR is a standard MODIS product (MOD15A2H), delivered at 500 m resolution and 8-day composites alongside LAI. The two products are retrieved jointly using a 3D RT look-up table approach.

Slide 20: Category C – Directional Radiative Quantities

Why Directional Quantities?

Satellite sensors observe canopy reflectance from a specific viewing direction — they do not integrate over the full hemisphere. The directional dependence of canopy reflectance (the BRDF) encodes rich information about canopy structure, leaf orientation, and even sub-pixel heterogeneity. Category C quantities characterize this directional behavior.

The Three Directional Quantities

- Hemispherical-Directional Reflectance Factor (HDRF): The ratio of canopy intensity exiting in direction Ω to the intensity from a perfect Lambertian (isotropic) reference panel, under combined solar and sky illumination.
- Bi-Directional Reflectance Factor (BRF): Same as HDRF but under direct solar illumination only.
- Bi-Directional Reflectance Distribution Function (BRDF): The fundamental physical quantity — ratio of exiting intensity to incident flux density, under mono-directional (solar only) illumination. Units: sr^{-1}.

Observational Context

HDRF is the quantity actually measured by field spectroradiometers when the sky is partially cloudy (i.e., mixed direct/diffuse illumination). BRF corresponds to clear-sky field measurements. BRDF is the model quantity that must be estimated through multi-angle satellite observations (e.g., MISR, POLDER, MODIS multi-angle compositing). The angular shape of the BRDF — bowl-shaped vs. dome-shaped — carries information about canopy structure (e.g., erectophile vs. planophile canopies).

Slide 21: Hemispherical-Directional Reflectance Factor (HDRF)

Definition

HDRF is defined as the ratio of intensity (radiance) exiting the canopy at the top in direction Ω to the intensity that would exit in the same direction from an ideal, perfectly reflecting, isotropic (Lambertian) reference panel under identical illumination conditions.

Formula

$HDRF_\lambda(\Omega, \Omega_0) = I_{s\lambda}(z=0, \Omega, \Omega_0) / I_{R\lambda}(z=0, \Omega, \Omega_0)$

- Numerator: Collided (scattered) intensity exiting the canopy upward in direction Ω at the top boundary.
- Denominator: Intensity from a Lambertian 100%-reflecting reference panel under the same illumination (both solar and sky).
- Both quantities are measured/computed under identical illumination — same solar position and same ratio of direct to diffuse irradiance.
- HDRF is dimensionless.

The Lambertian Reference Panel

A Lambertian (isotropic, conservative) white panel reflects all incoming radiation (no absorption) equally in all directions. Its radiance in any direction is $I_{ref} = (F_{0\lambda} + F_{d\lambda}) / \pi$, where the π in the denominator converts hemispherical flux to isotropic radiance. Normalizing by this reference removes the effect of illumination intensity, isolating the directional reflectance characteristics of the surface. In field measurements, spectralon panels serve as practical approximations to the Lambertian reference.

Slide 22: Bi-Directional Reflectance Factor (BRF)

Definition

BRF is the HDRF restricted to direct solar illumination only — the diffuse sky component is excluded. It is the most commonly modeled and retrieved directional reflectance quantity in remote sensing of vegetation.

Formula

$BRF_\lambda(\Omega, \Omega_0) = I_{s\lambda}(z=0, \Omega, \Omega_0) / I_{R\lambda}(z=0, \Omega, \Omega_0) = I_{s\lambda}(z=0, \Omega, \Omega_0) / [F_{0\lambda} \cdot (1/\pi)]$

- Under solar illumination only, the Lambertian panel radiance simplifies to $I_{R\lambda} = F_{0\lambda} / \pi$.
- Therefore: $BRF_\lambda = \pi \cdot I_{s\lambda}(z=0, \Omega, \Omega_0) / F_{0\lambda}$
- Illumination conditions for BRF: solar beam only, no diffuse sky — this is the 'black-sky' illumination condition, identical to DHR.
- BRF is dimensionless.

Relationship to BRDF

BRF and BRDF differ only by a factor of π: $BRF_\lambda = \pi \cdot BRDF_\lambda$. This factor arises from the normalization by the Lambertian panel radiance (which itself contains 1/π). In the literature, BRF is often the preferred quantity for field measurements and model comparisons because it is more interpretable — a perfectly Lambertian surface has BRF = 1 in all directions, regardless of its actual reflectance.

Slide 23: Bi-Directional Reflectance Distribution Function (BRDF)

Definition

The BRDF is the fundamental physical quantity describing the directional scattering properties of a surface. Unlike BRF and HDRF (which are dimensionless factors normalized by a reference panel), the BRDF has units of sr^{-1} and provides an intrinsic measure of how efficiently a surface redirects incident flux into each viewing direction.

Formula

$BRDF_\lambda(\Omega, \Omega_0) = I_{s\lambda}(z=0, \Omega, \Omega_0) / F_{0\lambda}(z=0, \Omega_0)$

- Numerator: Intensity (radiance) exiting the canopy upward in direction Ω.
- Denominator: Incident flux density in the solar direction Ω_0 — i.e., the direct solar irradiance $F_{0\lambda}$.
- Units: sr^{-1} (steradians inverse), because radiance is $W\ m^{-2}\ sr^{-1}\ nm^{-1}$ and flux density is $W\ m^{-2}\ nm^{-1}$.
- Only mono-directional (solar beam) illumination is considered — no diffuse sky.

Relationship to BRF

$BRDF_\lambda = BRF_\lambda / \pi$. For a perfect Lambertian surface with reflectance ρ, BRDF = ρ/π in all directions — a constant independent of viewing or illumination angle. Real surfaces have non-Lambertian BRDFs that vary with both illumination and viewing geometry.

MODIS and BRDF Retrieval

MODIS retrieves the BRDF by fitting a semi-empirical kernel-driven model (Ross-Li BRDF model) to multi-date, multi-angle surface reflectance observations. The MCD43 product suite delivers BRDF model parameters globally, enabling computation of BRF, BHR, DHR, and Albedo for any viewing/illumination geometry.